Currying, Partial Application & Composition

Season: 6 · FP Foundations · Category: Functional · Level: Intermediate

Intent

- Build pipelines from small functions.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Garden hose adapters.

When to Use

- Pure functional style in Java 25 + Zazr.
- The design force named in Intent shows up more than once.

When Not to Use

- Throwing exceptions everywhere is 'fine' for a script.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.functions;

import java.util.Map;
import java.util.concurrent.ConcurrentHashMap;
import java.util.function.Function;

public final class Fn {
    private Fn() {}

    public static <A, B, C> Function<A, Function<B, C>> curry(java.util.function.BiFunction<A, B, C> f) {
        return a -> b -> f.apply(a, b);
    }

    public static <A, B, C> Function<A, C> compose(Function<B, C> g, Function<A, B> f) {
        return a -> g.apply(f.apply(a));
    }

    public static <A, B> Function<A, B> memoize(Function<A, B> f) {
        Map<A, B> cache = new ConcurrentHashMap<>();
        return a -> cache.computeIfAbsent(a, f);
    }
}
```

Modern Java 25 / Functional

- Primary examples use Zazr; VAVR shown in comparison episode.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/functions/
```

Pitfalls

- Zazr rejects null inside Option/collections.
- Keep interpreters at the edge.

Interview Q&A

- Q: Interview: what problem does Currying, Partial Application & Composition solve? A: Build pipelines from small functions.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Build pipelines from small functions.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Garden hose adapters. [0:40] Intent — State the design force in one sentence: Build pipelines from s...